IoT-Enabled Modular Bird Repellent System: A Pattern-Based Approach for Enhanced Detection and Control

Stephen Eta Zhou and Neelam Mughees

***Abstract*—Effective bird repellent systems are crucial for mitigating the negative impacts of birds on agriculture, aviation safety, and urban infrastructure. This article presents an Internet of Things (IoT)-enabled modular bird repellent system that integrates real-time sensing, intelligent classification, and adaptive actuation to enhance efficiency and scalability. A pattern-based method is proposed for classifying birds and assessing the risk of bird populations. Experimental data showed that the detection accuracy of the perception module reached 95.2%, and the signal-to-noise ratio during daytime and nighttime was about 43.6 and 40.5 dB after stable image acquisition. The success rate of the driving module during the daytime within 500 m^2 was about 97.6%. The error rate of human–machine interaction module status detection was less than 2.6%, and the remote-control response time tended to be 157 ms. In communication testing, Ethernet, serial port, and USB communication had lower packet loss and error rates, and biomimetic bus's performance was better than conventional bus. After 48 h of on-site testing, the bird repellent efficiency reached 94.6%, with only two malfunctions and a recovery time of less than 5 s. The integration of functional modules ensured efficient collaboration, while the inclusion of multiple communication methods and a biomimetic bus enhanced the system's stability and adaptability to complex environments. This work demonstrates the potential of IoT-based intelligent bird repellent systems for precision wildlife management and environmental sustainability.**

***Index Terms*—Biomimetic bus, bird repellent system, modularization, pattern recognition, perception, software framework.**

I. Introduction

In THE era of rapid technological development, Internet of Things (IoT)-enabled systems are transforming various industries by enhancing automation, intelligence, and efficiency. Among them, IoT-driven bird repellent systems have special significance for certain industries, such as airports, agriculture, etc. [1], [2]. With the rapid development of aviation and agriculture around the world, conflicts between birds and human activities are becoming more frequent. Birds can cause significant damage to crop, leading to substantial economic losses for farmers. Similarly, bird strikes pose a serious threat to aviation safety, potentially causing catastrophic failures in aircraft. They also often interfere with solar panels, wind turbines, and other energy infrastructure, reducing efficiency and increasing maintenance costs. Many traditional bird control methods, such as nets, poisons, or physical traps, are harmful and unsustainable [3].

According to statistics, the global aviation economic loss caused by bird strikes is as high as billions of dollars every year [4]. In the field of agriculture, the invasion of crops by birds has also brought huge economic losses to farmers. For example, in some orchards and fields, birds peck at fruits and seeds, resulting in reduced or even extinct crops [5]. Therefore, the development of an efficient, intelligent and reliable bird repellent system has become a top priority. In recent years, with the rapid development of the IoT, artificial intelligence (AI), and machine learning (ML) technology, the design and implementation of the bird drive system has new technical means.

The IoT technology enables various sensors and devices to interconnect and realize real-time monitoring and data collection of bird information [6]. AI and ML algorithms can analyze and process the complex data collected to achieve accurate identification and prediction of bird behavior. The development of these technologies has provided strong support for the intelligent upgrade of the bird drive system. Therefore, there is a growing demand for the efficiency and intelligence of bird repellent systems.

In aviation, smart IoT-based bird repellent systems can significantly reduce the likelihood of bird strikes, ensuring passenger and aircraft safety. From the current research status, bird repellent technology has evolved from traditional simple physical repellent methods to more complex IoT-software driven intelligent systems [7]. Many studies are dedicated to using various sensor technologies to monitor bird activities, such as collecting data through sensors, such as sound and vision, and analyzing bird behavior patterns [8]. At the same time, there has been some progress in software control, with some systems being able to activate corresponding bird repellent devices based on preset rules [9].

However, there are still limitations in the IoT software design of existing bird repellent systems. The modularity of the software is insufficient, and the coupling between various functional modules is strong, resulting in poor maintainability and scalability of the system. When adding new features

Received 21 April 2025; accepted 15 May 2025. Date of publication 19 May 2025; date of current version 25 July 2025. This work was supported by the ShangHaiDieCheng-BM Software Unit, China. *(Corresponding author: Stephen Eta Zhou.)*

Stephen Eta Zhou is with the Software R&D Department, ShangHaiDieCheng-BM Software Unit, Shanghai 200000, China (e-mail: stephen.eta.zhou@outlook.com).

Neelam Mughees is with the School of Engineering and Technology, National Textile University, Faisalabad 37610, Pakistan (e-mail: neelam.mughees@ntu.edu.pk).

Digital Object Identifier 10.1109/JIOT.2025.3571532

or improving existing ones, it is often necessary to make significant modifications to the entire software system. In addition, the application of pattern-based methods in design is not mature enough, and there is a lack of unified design pattern guidance, which makes the software less flexible in dealing with complex bird behavior and changing environmental scenarios, making it difficult to achieve efficient and accurate bird repellent effects. Therefore, the research will focus on advanced technologies in IoT software framework design, with bird repellent system software as the object. The aim is to solve the problems in existing bird repellent system software design and improve its performance and adaptability. The innovation of the research needs to be reflected in various strategies, including dynamic task allocation, biomimetic bus design, etc., to ensure the high efficiency and system stability of bird repellent.

II. Background

This literature review aims to evaluate recent developments in IoT-enabled bird repellent system design, particularly focusing on software innovations, software optimization, and adaptability enhancements. Despite advances in bird repellent technology, existing systems often suffer from limited adaptability, inconsistent performance, and suboptimal software design. Cho et al. [10] addressed the development of an acoustic deterrent device for protecting fishponds and other areas from unwanted bird presence. The system aimed at improving repelling performance through targeted sound emissions.

Li and Yu [11] proposed an algorithm designed to prevent bird collisions by identifying two major categories from representative small birds and airplanes, utilizing a joint attention mechanism to enhance detection accuracy. Another study detailed the development of a deep-learning-based model for bird detection, from its creation to deployment in an RP2040 microcontroller, aiming to detect pest bird sounds and estimate their proximity [12]. Similarly, another study [13] presented an automatic bird repellent system utilizing deep learning for bird detection, combined with a laser mechanism for deterrence. The system effectively reduced wild bird presence in agricultural settings, demonstrating a daily repulsion rate of 40.3%.

The development of software frameworks aims to provide efficient and reliable solutions for problem-solving in different fields by building software architectures with specific functionalities, promoting technological progress and innovative development in various industries. In [14], a specific, controlled automotive network security testing environment has been provided for researchers and engineers to quickly improve their skills. This framework simulated a bus controller network, tested attack vectors and mitigation methods, and provided basic implementation for common attack types. Alonso et al. [15] proposed a hybrid meta framework to analyze and evaluate the security and privacy issues of container orchestration platforms, in order to help small and medium-sized manufacturing companies better adapt to Industry 4.0. This framework referred to recognized security and privacy standards and frameworks, which could understand assets and requirements, select and configure countermeasures.

Gemeno [16] elaborated on the design framework for predicting software defects, as well as the scientific methods and data sources used. The big data analysis for defect segmentation made significant contributions to the field of predicting defects. The research incorporated these algorithms and implementations as part of ML applications to enhance product quality and gain competitive advantage. Sha et al. [17] proposed a software–hardware collaborative layered memory management framework to address challenges, such as performance isolation, context switching, and resource overuse in virtualization. Based on the unique system functionality in virtualization, this framework automatically determined page popularity and migrated pages between fast and slow memory for better performance. The experimental results showed that this framework outperformed existing hierarchical memory management designs in virtualization.

An open-source quantum ML framework written in Python was proposed in [18] that assisted researchers in conducting experiments, with a particular focus on quantum kernel technology. This framework implemented multiple state-of-the-art algorithms, analyzed data through quantum kernels, which could be used as a library to integrate into existing software, improving code reuse rates. Software optimization is based on software development, which improves key indicators, such as software performance, maintainability, and security through algorithm improvement, architecture adjustment, and resource management, in order to achieve better operational results and user experience [19].

A TreeViewer software developed in [20] was used to draw phylogenetic trees. This software had flexibility, modularity, and user friendliness, which easily generated highly customized and publishing quality tree diagrams. In addition, TreeViewer could run for free on Windows, macOS, and Linux operating systems. Picone et al. [21] proposed a new edge digital twin architecture model and its implementation method, which effectively performed lightweight replication of physical devices and provided autonomous and standard collaborative digital abstraction layers for things and services. Experimental analysis showed that this method exhibited good performance in multiple IoT application environments and had significant advantages compared with existing advanced methods.

Another study [22] compared the impact of event driven architecture and Rest style on modularity through five evolutionary scenarios. The results showed that the event-driven architecture performed better in separating focus points. Overall, the Rest style performed better in metrics, such as coupling, cohesion, complexity, and scale. AI methods were used to explore the data sources of the software development industry, software ecosystems, and software system customers in [23] in order to proactively address security vulnerabilities. By constructing an intelligent knowledge management system and applying ML and deep learning methods, software security vulnerabilities could be effectively managed, providing insights for software development activities.

The performance of five different heuristic algorithms on the optimal clustering problem of software modules through experiments was compared in [24]. The bat algorithm, cuckoo algorithm, and teaching-based algorithm performed better than other algorithms. In addition, the study also found that when using logistic chaos method to generate the initial population, the clustering quality of these algorithms was 3.155, 3.120, and 2.778, respectively.

Another study [25] introduces an IoT-based bird prevention device designed to mitigate issues like false alarms and system instability prevalent in traditional systems. By integrating deep learning, specifically convolutional neural networks (CNNs), the system enhances bird image recognition accuracy. The incorporation of multisensor fusion and real-time data transmission bolsters the system's intelligence and remote monitoring capabilities. Results indicated significant improvements in bird detection accuracy, recall, and overall system stability, with notable reductions in false alarm rates and processing times.

Phetyawa et al. [26] examined the implementation of IoT-based bird repellent systems, emphasizing the challenges posed by bird behavior. It discussed the necessity for adaptable systems capable of responding to varying bird patterns and behaviors. The study underscored the importance of integrating behavioral analysis into the design of repellent systems to enhance their effectiveness and adaptability.

Similarly, Adami et al. [27] presented an intelligent animal repelling system utilizing embedded edge-AI for crop protection. The system employed computer vision and ultrasound emission to detect and deter animals, with real-time object detection models (YOLO and Tiny-YOLO) deployed on edge computing devices like Raspberry Pi and NVIDIA Jetson Nano. The research highlighted the system's capability to operate efficiently in rural environments with limited energy supply and network connectivity, demonstrating a balance between performance, cost, and energy consumption

A. Research Gaps and Contributions

In summary, although existing software frameworks and optimization methods have achieved significant results in different fields, there are still some shortcomings. The modularity of the software is insufficient, and the coupling between various functional modules is strong, resulting in poor maintainability and scalability of the system. Moreover, the design framework for software defect prediction still needs further optimization to improve prediction accuracy. In terms of software optimization, balancing indicators, such as focus separation and coupling, remains a challenge.

In addition, the application of pattern-based methods in design is not mature enough, and there is a lack of unified design pattern guidance, which makes the software less flexible in dealing with complex bird behavior and changing environmental scenarios, making it difficult to achieve efficient and accurate bird repellent effects. Therefore, this study has proposed a modular and pattern-based approach to software design, aiming to better address issues in software design and improve software reliability and maintainability.

First of all, the system is divided into man–machine interface, data communication, bird target confirmation module, data processing module, and other key functional modules to improve the maintainability and scalability of the system. Second, the pattern-based method is used to process bird information, and the fuzzy matching algorithm is used to realize the classification and identification of birds and the assessment of the risk degree of birds, so as to improve the flexibility of the system to deal with the complex environment. These key steps and methods lay the foundation for the follow-up research work.

The research has focused on advanced technologies in software framework design, with bird repellent system software as the object. The results have solved the problems in existing bird repellent system software design and improved its performance and adaptability.

B. Article Organization

The remainder of this article is organized as follows. Section II introduces the design methodology of the bird repellent system software, covering its architecture and software structure. Section III discusses the implementation and testing results of the software system, providing a detailed analysis. Finally, Section IV concludes this article by summarizing the effectiveness of the proposed modular and pattern-based approaches.

III. Design Method of Bird Repellent System Software

A. IoT Software System Framework Design

In modern agriculture and engineering, bird repellent systems have become a valuable tool for preventing birds from causing damage to crop and facilities. To improve the performance of bird repellent systems, this study introduces modularization and pattern-based methods to make the design of bird repellent systems clearer, more efficient, and flexible, to meet development needs. The bird repellent system aims to improve the efficiency of bird repellent and regional safety management, including functions, such as information collection, organizational scheduling, target confirmation, automatic repelling, and comprehensive control [28], [29].

Information collection is used to gather information on bird species, quantity, activity time, and other related factors. Organizational scheduling is used to coordinate the equipment, such as radar and audio systems. Target confirmation is used to identify birds and determine their location and quantity. Automatic dispersal is conducted through sound and light interference, visual stimulation, etc. Comprehensive control is used to control equipment on/off, parameter settings, etc. [30]. The system integrates various modules and uses ML, image processing, and other technologies to monitor bird activity, issue real-time warnings and take measures to drive away, while collecting data for analysis to provide decision-making basis for management departments and optimize workflow [31]. The software structure of the bird repellent system is shown in Fig. 1.

As shown in Fig. 1, the system consists of four parts. The human–computer interaction interface includes interface display, user input, data transmission and reception, and

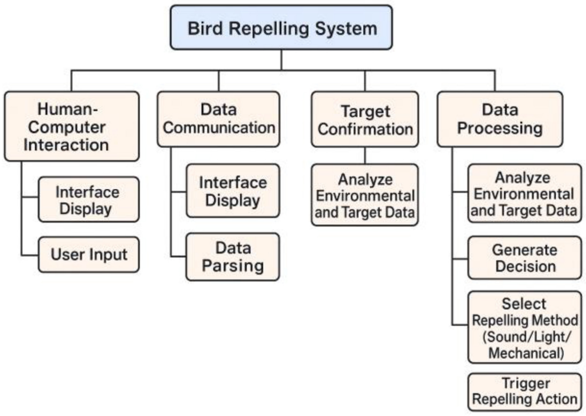

Fig. 1. IoT software structure of the bird repellent system.

data parsing, which is mainly responsible for the interaction between users and the system [32]. Through the interface display, users can intuitively see the operation status of the system. User input allows users to input operational instructions. Data transmission and reception are responsible for exchanging data with external parties. Data parsing processes the received data, so that it can be understood by the system.

Data communication plays a role in connecting various modules, and the bird target confirmation module is mainly responsible for identifying and determining bird targets within the airport area [33], [34]. The data processing module includes the generation of bird repellent decisions, analyzing the data provided by the bird target confirmation module, and generating corresponding bird repellent strategies to achieve effective bird repellent goals. The architecture of the bird repellent system is shown in Fig. 2.

The bird repellent system architecture is divided into five layers. The top layer is the business layer, which includes product management and analytical decision-making. Functional modules are mainly distributed in the application layer and the service layer. The second layer is the application layer, which contains the man–machine interaction module, which is used to realize the interactive operation between the user and the system. The third layer is the service layer, including the intelligent perception module and the intelligent drive away module, responsible for the identification and confirmation of bird targets and the generation and execution of bird drive decisions. The fourth layer is the control layer, which includes positioning and bird driving command control, equipment status acquisition, and communication. Positioning and bird driving command control includes driving the industrial computer. The equipment status acquisition and communication includes the information industrial computer, communication equipment, and display terminal. The bottom layer is the physical layer, which includes physical devices, such as radar, laser, bird repellent gun, strong sound, and optoelectronics [35].

The hardware components used in the system include a variety of sensors (such as optical detection units, laser detection units, radar calibration units, and infrared detection units), actuators (such as stroboscopic control devices, laser control devices, and audio control devices), and microcontrollers.

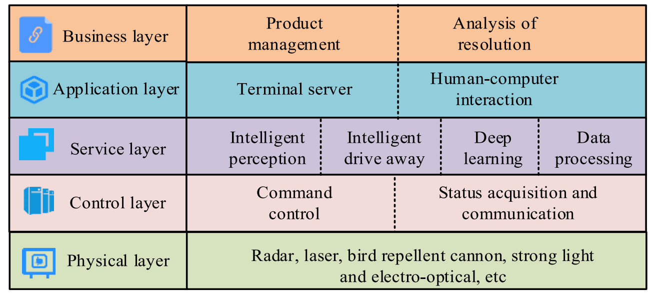

Fig. 2. Bird repellent system architecture.

The sensor is responsible for collecting bird information and converting it into electrical signals, which are preliminarily processed by the microcontroller and sent to the software system. The software system generates bird repelling instructions according to the processing results and controls the actuator through the microcontroller for bird repelling operation. The close cooperation between hardware and software ensures the efficient operation of the system.

B. Functional Module Design

As shown in Fig. 3, the functional modules of the bird repellent system consist of four main modules, each with clear division of labor and unique functions, working together to achieve efficient bird repellent and stable system operation. The perception module is the "eye" of the system, which includes optical detection unit, laser detection unit, radar calibration unit, infrared detection unit, and multisource detection control turntable. Through various principles, such as optics, laser, radar, infrared, and precise control of the turntable, the presence of birds can be perceived from all directions and angles, providing accurate information for subsequent processing and dispersal work.

The information processing module is like the "brain" of the system, in which the comprehensive management unit plays a core coordinating role, responsible for coordinating the work of other sub modules. The information processing unit conducts in-depth analysis and processing of the detected data, while the image acquisition unit is responsible for collecting detection images. The wireless communication unit can achieve fast data transmission, ensuring smooth information flow throughout the entire system [36].

The drive module is the "weapon" of the system, equipped with a universal working turntable, strobe control equipment, laser control equipment, and audio control equipment. The universal work turntable provides support and controls the direction of the bird repellent device. The strobe control device emits strong flashes, the laser control device uses laser beams, and the audio control device plays specific sounds to repel birds from visual, energy, and auditory aspects, achieving good bird repellent effects.

The human–computer interaction module is the "bridge" between the system and the operator, including a status monitoring unit, a comprehensive display unit, an intelligent operation unit, and a remote-control unit. The status monitoring unit monitors the real-time operation status of the

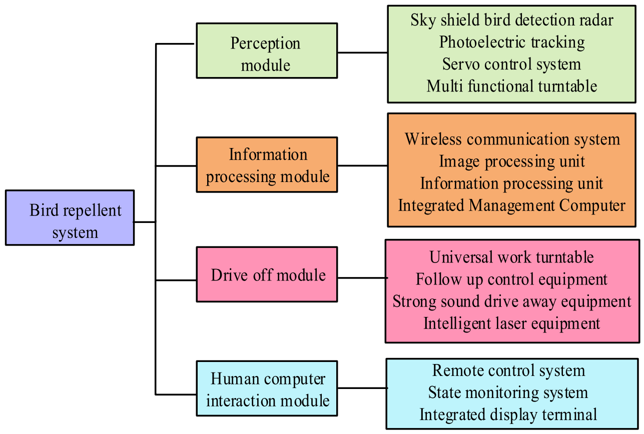

Fig. 3. Bird repellent system function module.

system to ensure its stability and safety. The integrated display unit can intuitively display various information of the system, making it easy for operators to understand. The intelligent operating unit provides convenience for system operations, making them simpler and more efficient. The remote control unit allows operators to perform remote control and monitoring, greatly improving the flexibility and operability of the system [37]. These four modules and their submodules work closely together to enable the bird repellent system to efficiently detect, process, and drive away birds, while ensuring the stable operation and convenient operation of the system.

C. Dynamic Task Allocation Strategy

The dynamic task allocation strategy of the bird repellent system is a critical task that involves many factors to ensure effective removal of birds from dangerous areas to safe areas. This strategy uses a priority-based algorithm to determine the priority of each task. The priority of tasks is determined based on their urgency and dangerousness [38]. The urgency level refers to the length of time birds spend in a hazardous area, while the danger level refers to how far away birds are from the edge of the hazardous area. Based on these factors, tasks are assigned a priority value, so that the system can prioritize processing the most urgent and dangerous tasks. This strategy also considers device dynamic resource constraints, including the current state of each device, to determine which tasks each device can handle.

The system ensures that each task is assigned to a device with sufficient resources to complete the task, thereby avoiding task failures caused by device overload or insufficient resources [39]. In order to achieve efficient dynamic task allocation, this strategy uses real-time data to update the status of tasks and the resource status of devices. When birds are driven away to a safe area, the task status will be updated to "completed." If the task fails or the device is unable to complete the task, the system will reassign the task to ensure its completion.

The bird repellent device can undertake multiple tasks in sequence when processing tasks but can only complete the expulsion of one main bird flock at a time, and the task needs to be driven to a safe area before it ends. Its tasks are allocated based on different levels of urgency and priority to ensure high priority processing. The processing server has sufficient computing power to simultaneously handle multiple tasks to meet real-time requirements. Different tasks are independent of each other, and device execution does not interfere with each other, which is conducive to resource allocation and scheduling. The set of bird repellent devices is shown in

$$\begin{cases} Dev = \{r_1, r_2, \dots, r_n\} \\ r_i = \{id_r, tas_r, pos_r\}. \end{cases} \tag{1}$$

In (1), the set of devices is Dev, the number of devices is n, the device number is id_r, the number of received tasks is tas_r, and the bird repellent distance of the device is pos_r. The set of bird repellent tasks is shown in

$$\begin{cases} Tas = \{t_1, t_2, \dots, t_m\} \\ t_j = \left\{id_t, tim_g, tim_w, tim_d, pri_i, pri_d, rsi, saf\right\}. \end{cases} \tag{2}$$

In (2), the number of bird species to be driven is m, the task number is id_t, the task production time is tim_g, the task waiting time is tim_w, the task end time is tim_d, the initial priority of the task is pri_i, and the dynamic priority of the task is pri_d. The bird risk location is rsi, and the safe evacuation location is saf. The equipment efficiency is shown in

$$\eta = \frac{\mathrm{dis}_l}{di_l s + \mathrm{dis}_e}. \tag{3}$$

In (3), the equipment efficiency is η, the load distance is dis_l, and the no-load distance is dis_e. The objective function for task allocation is shown in

$$\begin{cases} f = \min\left[\left(\mathrm{dis}_t^i - \min\left(\mathrm{dis}_t^j\right) + \varphi \times \eta\right)\right] \\ \mathrm{dis}_t^j = \mathrm{dis}_{lt}^i + \mathrm{dis}_{et}^i \end{cases}. \tag{4}$$

In (4), the total distance traveled by device r_i in task t_j is dis_t^j. The load distance is φ. The empty distance is dis_{et}^i. The efficiency coefficient is φ. The task time constraint is shown in

$$tim_d^j > tim_g^j + tim_w^j. \tag{5}$$

In (5), the end time of task t_j is tim_d^j, its start time is tim_g^j, and its waiting time is tim_w^j. The objective function for bird repellent task allocation focuses on considering the rotation distance and dispersal efficiency of the equipment executing the task, ensuring that the task is completed before the specified time, and ensuring that each task has corresponding equipment processing to drive the bird flock away to a safe area. The bird repellent task allocation strategy determines the execution order of bird repellent tasks.

Common strategies include first-come–first-served, priority strategy, and shortest task priority. Based on a static priority strategy, task priority is determined and remains unchanged during creation, which has advantages in specific scenarios, but may cause problems when there are multiple tasks and fewer devices. Therefore, considering the initial priority and waiting time of tasks, a dynamic priority strategy is proposed. The initial priority division of tasks is shown in

$$pri_i \in \{p_1, p_2, p_3, p_4, p_5, p_6, p_7, p_8, p_9, p_{10}\}. \tag{6}$$

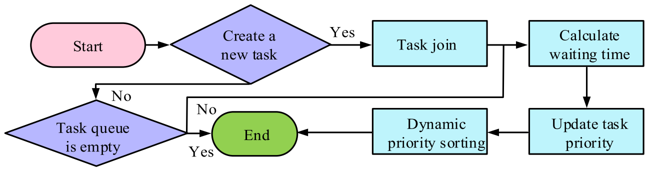

Fig. 4. Dynamic priority allocation process.

In (6), the initial priority includes ten levels. The initial priority can reflect the importance of the task at the time of creation. Tasks with higher initial priority will have higher dynamic priority, which allows tasks that are initially identified as important to maintain a relatively high priority level throughout the entire execution process [40]. At the same time, dynamic priority will also increase based on the waiting time of tasks. The longer the waiting time, the faster the improvement speed. The dynamic priority calculation is shown in

$$\begin{cases} pri_d^j = \dfrac{\left[pri_i^j\left(pri_i^j-1\right) + T_w^j\left(T_w^j-3\right)\right]}{2} + pro_i^j T_w^j + 1 \\ T_w^j = \frac{10tim_w^j}{tim_d^j - tim_g^j}. \end{cases} \tag{7}$$

In (7), the dynamic priority of task j is pri_d^j, and its initial priority is pri_i^j. The relative waiting time for the task is T_w^j, the waiting time is tim_w^j, the start time is tim_d^j, and the end time is tim_g^j. The dynamic priority allocation process is shown in Fig. 4.

The process starts from the beginning. The system first checks whether a new task has been created. If there are no new tasks, the system will further check if the task queue is empty. If the task queue is empty, the process ends here. If the task queue is not empty, subsequent operations are continued. When a new task is created, it will be added to the task sequence. Afterwards, the system calculates the waiting time of the task to be driven at the current time. Based on this waiting time, the system will update the priority of the task. Finally, all tasks are sorted based on the updated dynamic priority. The entire dynamic priority allocation process is completed.

D. System Software Operation Process

The software of the bird repellent system adopts a multithreaded running mode, which has many advantages, such as low cost, short switching time, and convenient communication. In its workflow, interactive data exchange between each module is achieved through shared data areas, ensuring efficient collaboration among modules [41]. The multithreaded running mode ensures that the system can handle multiple tasks simultaneously, improving overall running efficiency and response speed. By sharing data areas for data exchange between modules, not only does it simplify the process of data transmission, but it also enhances data consistency and reliability. The system operation process is shown in Fig. 5.

In Fig. 5, after the system is started, the bird repellent system is first initialized. After initialization is completed, the system will perform four operations simultaneously, namely,

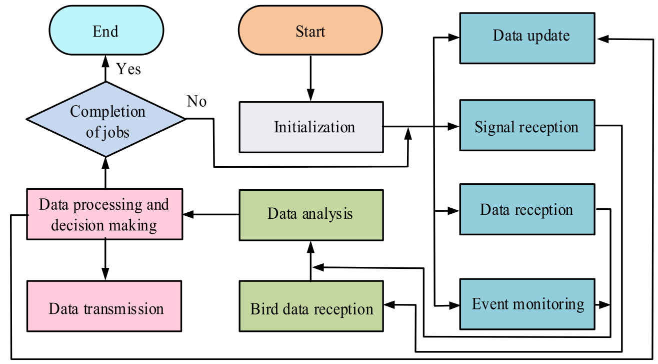

Fig. 5. System operation process.

data refresh of the bird repellent system, clock signal reception, bird repellent data reception, and event monitoring, such as clicking. After receiving the data, the system receives bird data through UDP and then parses the bird data. The parsed data will be processed or generate bird repellent decisions. Afterwards, the system will send UDP and serial data. During the entire operation process, the system will continuously check whether to end the work. If the result is yes, the system will shut down. If the judgment result is negative, the system will continue to run the above process. The system workflow is shown in Fig. 6.

In Fig. 6, the system first performs a system status check after startup and then enters working mode or inspection mode. In working mode, there are two options: 1) automatic mode and 2) manual mode. In automatic mode, after capturing the target through radar, the target information that needs to be issued is selected, and then automatically guided the electro-optical equipment to find the target. In manual mode, the photoelectric device is manually guided to find the target. Whether in automatic or manual mode, the device will determine whether to track the target after finding it.

If tracking is required, bird repellent decision selection and target forward prediction are carried out, followed by turning on directional sound and laser drive to deal with the target. Then, it is determined whether there is still a target. If so, the above operation is repeated. If no tracking is required, reach the designated position or rotate at a constant speed. In inspection mode, the entire system workflow is completed by operating recognition devices, such as photoelectric and radar, as well as directional sound, laser and other driving devices, and finally uploading inspection logs [42].

E. Design of Biomimetic Bus Based on Patterns

The complexity and diversity of bird information have brought many challenges to system design, and pattern-based biomimetic bus design provides an effective way to solve this problem. This design mainly uses a unique fuzzy matching algorithm to process various feature information of birds, including key elements, such as body length and speed, in order to achieve classification and recognition of birds. In addition, it also uses the weighted average method to calculate the danger level of bird flocks, fully considering the different weights of factors, such as bird size and speed in

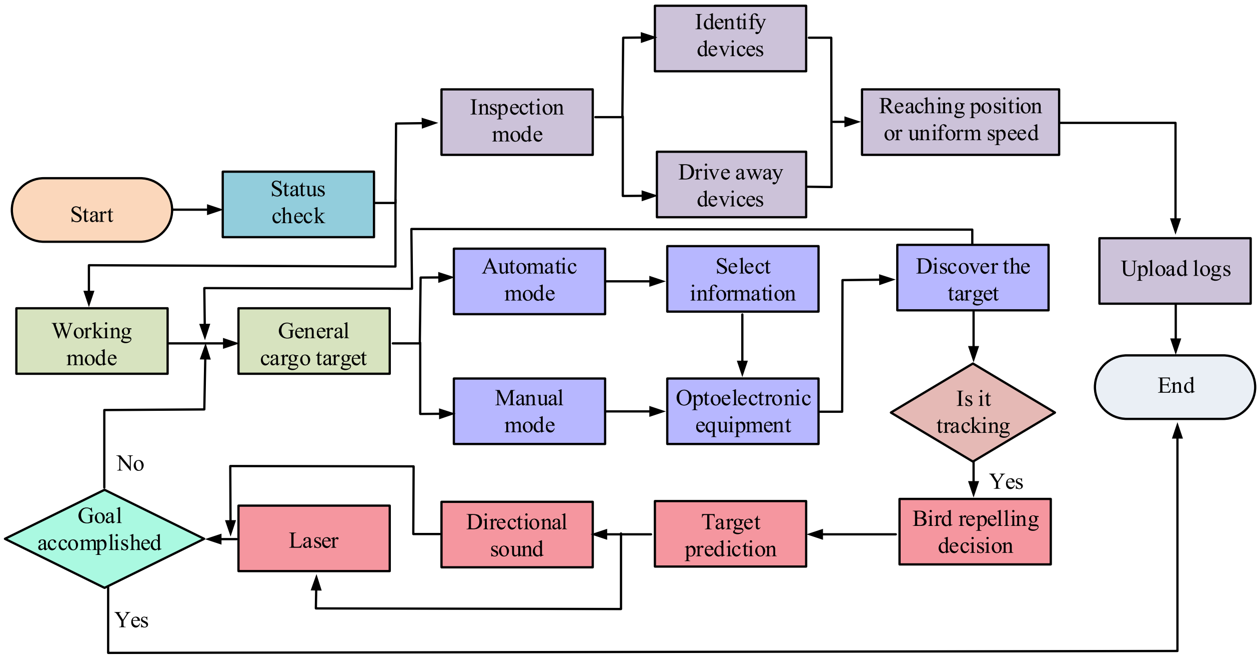

Fig. 6. System workflow.

the calculation process, making the assessment of bird flock conditions more scientific and accurate.

The reason of selecting the fuzzy matching algorithm is that the information of birds is complex and diverse, and the characteristic information, such as body length and speed, has some fuzziness and uncertainty. Fuzzy matching algorithm can quantitatively process these fuzzy information through membership function, so as to realize the classification and identification of birds and the accurate assessment of the risk degree of birds, and provide scientific basis for the decision of bird repellent.

The bird membership degree is shown in

$$D = e^{\left(-0.5z\left(\frac{l-y}{\sigma}\right)^2\right)} \tag{8}$$

where the bird membership degree is D, the body length is represented as l, the center value is y, the standard deviation is σ, and the wingspan length of the bird is z. The danger factor for bird flocks is shown in

$$\mathrm{Dan} = \frac{\sum_{i=1}^{m} \alpha_i S_i}{\sum_{i=0}^{m} \alpha_i} + \frac{\sum_{i=1}^{m} \beta_i V_i}{\sum_{i=0}^{m} \beta_i} \tag{9}$$

where the size of the bird is represented as S_i, and its weight proportion is α_i. The speed of birds is represented as V_i, and its weight proportion is β_i. The communication module is responsible for controlling the data transmission between the computer and various devices, covering data analysis and packet transmission. It adopts communication methods, such as Ethernet, serial port, USB, etc., to meet different requirements. The network topology structure is shown in Fig. 7.

According to relevant standards, Ethernet technology standards for communication networks have different speeds for network cards. For physical connection topology, although bus topology has certain characteristics, such as short cables and ring topology, star topology is chosen due to its high scalability and short network latency, with Ethernet switches as the central node. TCP and UDP communication protocols are commonly used in Ethernet communication, and UDP/IP protocol is adopted in research due to its good data transmission efficiency. When using UDP communication for bird repellent devices, there is no need to establish a connection with the server.

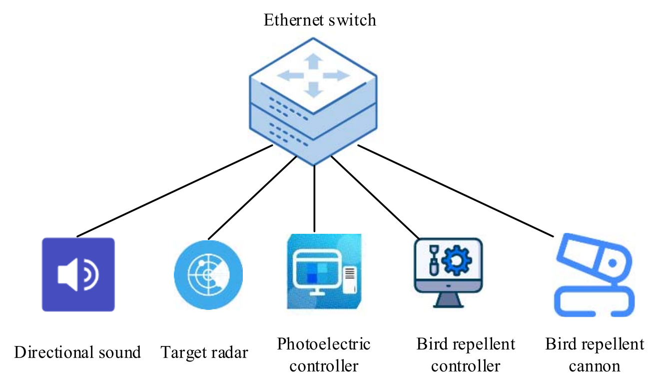

Fig. 7. Network topology.

Data transmission and reception are achieved through specific processes. Different devices have different communication methods, and data has specific formats and transmission rates. Each module can interact with data and determine data integrity and correctness through check bits. Communication protocols need to be developed for data parsing, packaging, and transmission, and the system processes data according to the protocol. In addition, the redundant configuration of system equipment includes primary backup switching, hot backup, cold backup, and other methods.

The bird repellent device has two sets of parallel active parts and dual computer systems and can also be designed with dual bus redundancy. At the same time, there is software information redundancy for functions, including data backup and redundant computing, to ensure stable system operation.

IV. Software System Implementation and Testing

A. Experimental Environment Setup

In terms of software system implementation, the Launch file function in the robot operating system is utilized to incorporate relevant launch files, including the launch function node manager, loading recognition camera devices, and control devices. A user-friendly intelligent bird repellent interface is developed using the C++ application development framework, and the program is written in Python language.

In the service layer of the Floudian system, the deep learning components mainly include convolutional neural network (CNN) for image processing and object detection, and recurrent neural network (RNN) for time-series analysis of bird behavior patterns. The data preprocessing component includes data cleaning, normalization, and feature extraction modules to ensure the data quality and accuracy of the deep learning model input. The cooperative work of these components provides strong support for the efficient operation of the system. In terms of hardware device access, laser gimbal, strong sound gimbal, and photodetector-related hardware devices are connected to the system in the laboratory.

During system testing, the calculation function diagram is opened to confirm that all required nodes are open and topic communication is running normally. The bird repellent software interface can display various status information of the current device. In the minimum communication clock cycle test of the system, 20 ms is used as the minimum unit in the communication cycle. The timing signal generator generates the 20-ms timing signal, and the bird control equipment and upper computer in the airport bird control system receive the 20-ms timing signal uniformly. The optoelectronic equipment and laser use 200 ms as the communication cycle. The algorithm training and prediction process is implemented in the Python environment of the Windows system, built on the Pytorch framework. The computer equipment is mainly configured with Intel Core i9 10920X CPU and NVIDIA TITAN RTX GPU.

The parameters of the fuzzy matching algorithm are set as follows: the central value of the membership function is determined according to the actual measurement data of bird body length and wingspan length, and the standard deviation is obtained by statistical analysis of the distribution range of bird characteristic data. When calculating the risk factor of birds, the weight ratio of bird size and speed is set to 0.6 and 0.4, respectively, and these parameter values are based on the comprehensive analysis of bird behavior patterns and bird repelling effects.

B. Functional Testing of Each Module

The test results of the perception module are shown in Fig. 8. Fig. 8(a) shows the accuracy test results. As the number of tests increased, the accuracy fluctuated. The detection accuracy reached 95.2%. It performs stably in different environments and bird species. Fig. 8(b) shows the response time test results. As the number of tests increased, the response time gradually decreased and eventually approached 215 ms. The results indicate that the perception module performs well in testing, with high accuracy and fast response speed, which can meet real-time requirements. Meanwhile, the perception module may be influenced by other factors, including sensor quality, environmental factors, algorithm parameters, etc. Therefore, in practical applications, it is necessary to comprehensively consider these factors to ensure the performance and stability of the perception module.

The test results of the information processing module are shown in Fig. 9. Fig. 9(a) shows the test results of data processing speed. As the amount of data increased, the data processing time gradually increased. When the data volume was 1000, the processing time was approximately 2567 ms. Fig. 9(b) shows the quality of image acquisition, which gradually increased with the increase of image quantity and eventually stabilized. After stabilization, the signal-to-noise ratio of images collected during the daytime and nighttime was approximately 43.6 and 40.5 dB, respectively. The results indicate that the data processing speed slows down with the increase of data volume, while the image acquisition quality first increases and then stabilizes with the increase of image quantity. The reason is that the increase in data volume increases the processing burden and time, while the increase in the number of images ensures stable quality after system optimization.

The test results of the drive module are shown in Fig. 10. In Fig. 10(a), the displacement range was 500 m^2. After ten tests, the successful drive rate during the daytime was over 95%, and the average successful drive rate was about 97.6%. The success rate at nighttime was over 93%, and the average successful drive rate was about 94.2%. In Fig. 10(b), as the displacement range increased, the successful drive rate gradually decreased. When the displacement range exceeded 950 m^2, the successful drive rate dropped to about 90%, with a daytime displacement rate of about 90.1% and a nighttime displacement rate of about 87.6%. This indicates that the drive module has a good displacement effect within a certain range, but as the range expands, the displacement effect decreases. The performance of the drive module is significantly affected over a large range due to factors, such as power limitations and signal propagation attenuation.

The test results of the human–computer interaction module are shown in Fig. 11. Fig. 11(a) shows the accuracy of state detection, with an error rate of less than 2.6% compared with the actual system operating state. Fig. 11(b) shows the response time of remote control. As the number of tests increased, the response time gradually decreased and eventually stabilized. When the number of tests exceeded 30, the response time of remote control was 157 ms. The results indicate that the state detection accuracy and response speed of the human–computer interaction module are high, which can meet the practical application requirements.

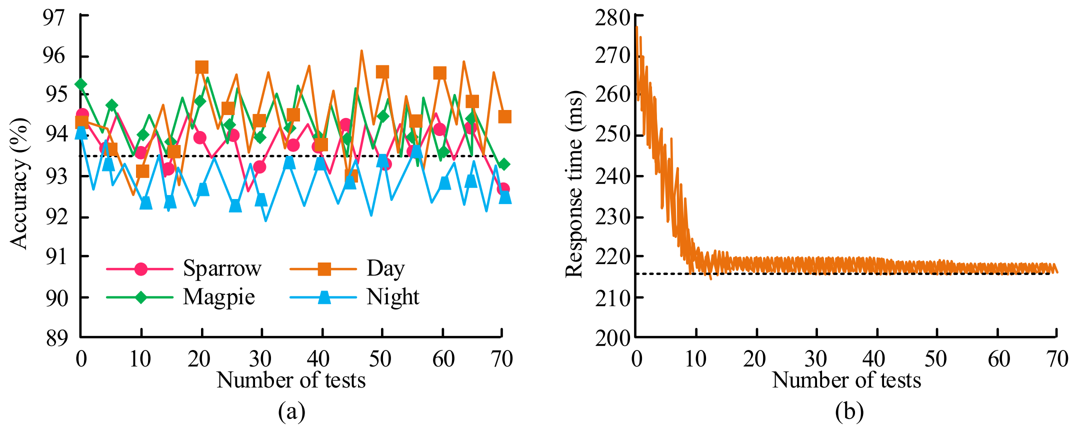

Fig. 8. Test results of the perception module. (a) Accuracy. (b) Response time.

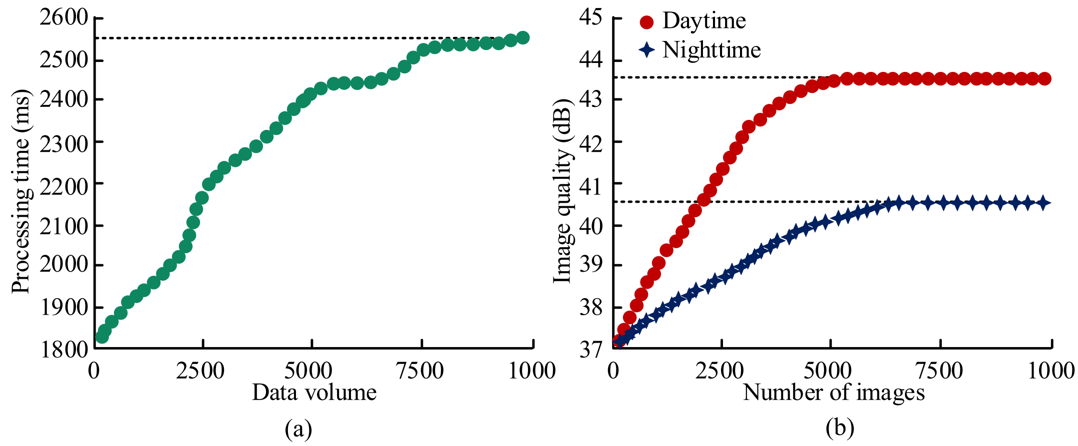

Fig. 9. Test results of the information processing module. (a) Processing time. (b) Image quality.

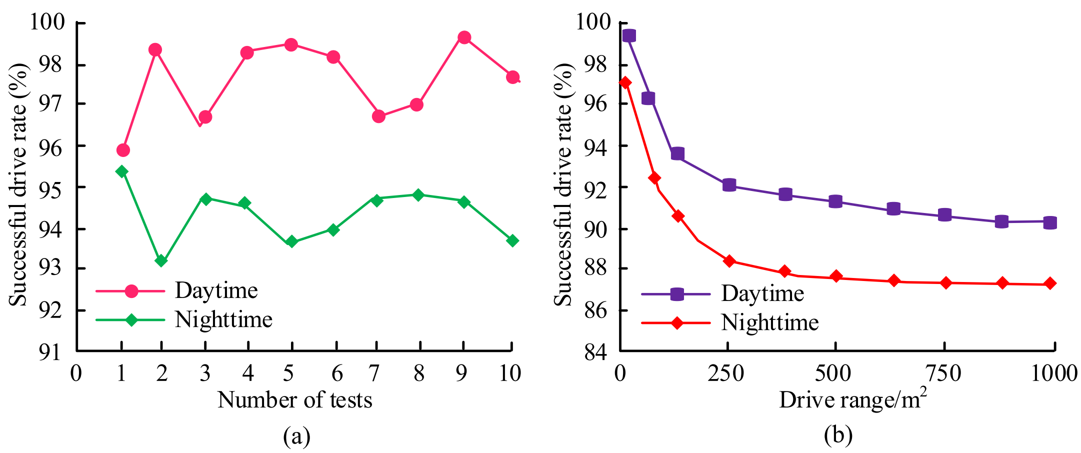

Fig. 10. Testing results of the drive module. (a) Displacement range is 500 m^2. (b) Increased displacement range.

C. Communication Test Results and Discussion

During the communication testing process, the communication bandwidth and latency are shown in Fig. 12. Fig. 12(a) shows the results of communication bandwidth testing. The average Ethernet communication bandwidth reached 1000 Mb/s, the average serial communication bandwidth was 115.2 kb/s, the average USB communication bandwidth was 5 Gb/s, and the average USB communication bandwidth was 480 Mb/s. Fig. 12(b) shows the communication latency testing results, with an average Ethernet communication latency of 5 ms, a serial communication latency of 20 ms, and a USB communication latency of 3 ms.

The results indicate that in the communication testing of the bird repellent system, the communication bandwidth between USB and Ethernet is much larger than that of the serial port, demonstrating the efficient communication capability of

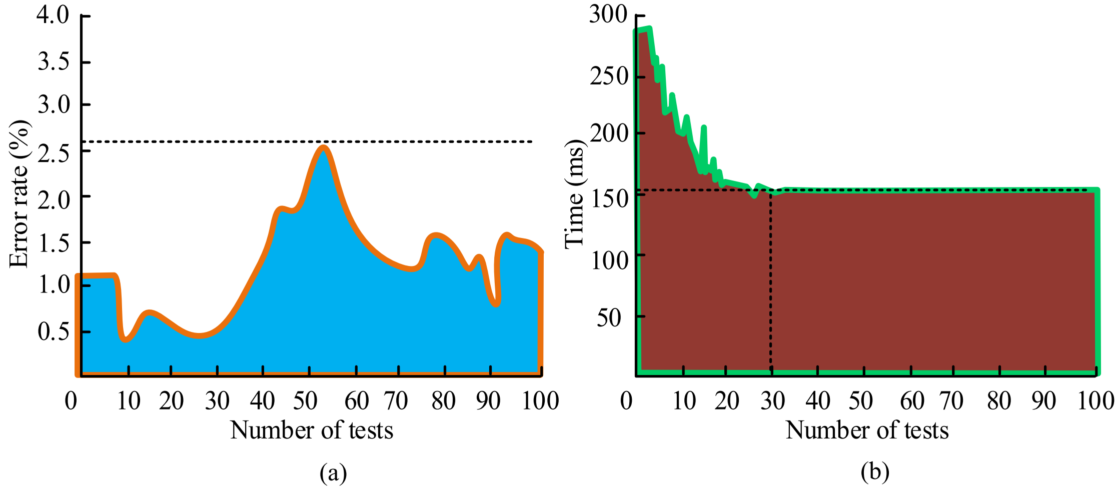

Fig. 11. Testing results of the human–computer interaction module. (a) Accuracy of state detection. (a) Response time of remote control.

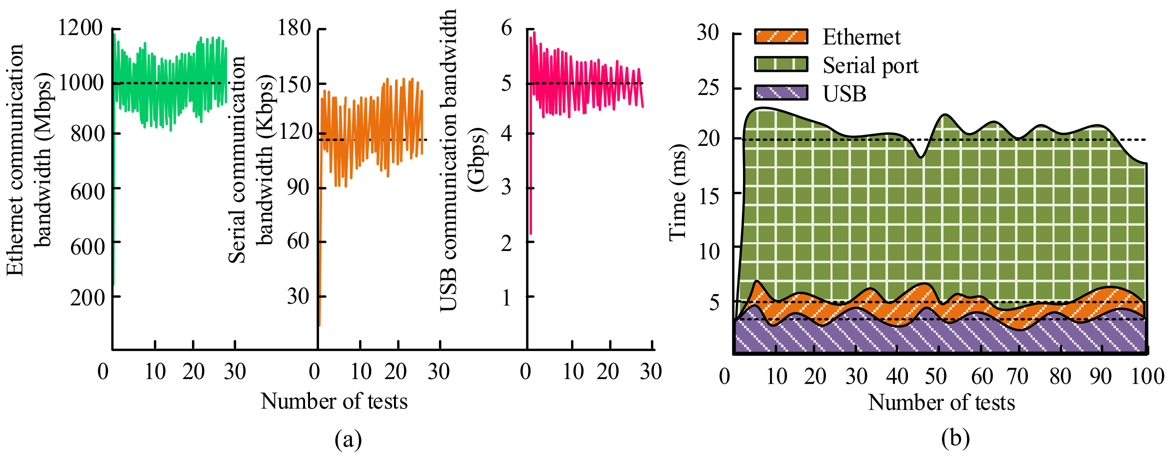

Fig. 12. Communication bandwidth and latency. (a) Bandwidth. (b) Communication latency.

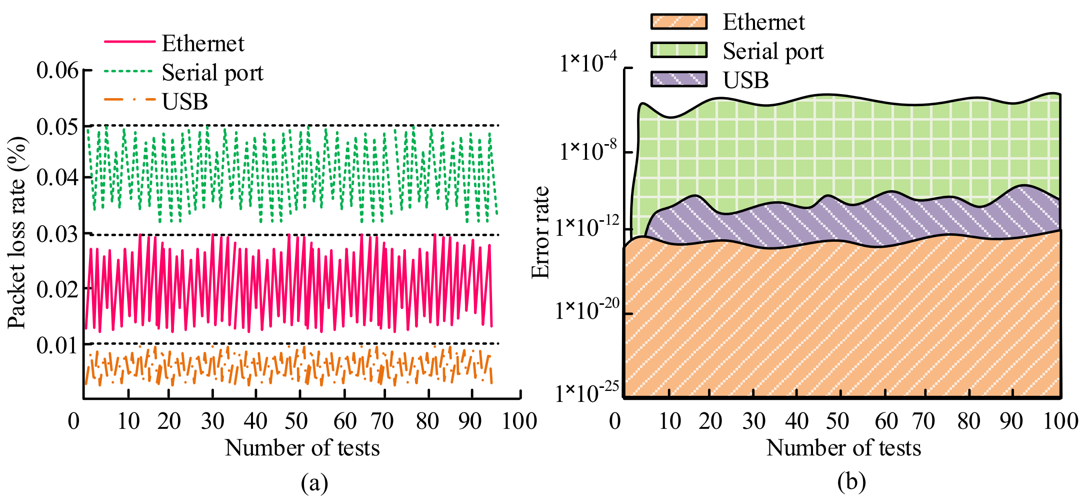

Fig. 13. (a) Packet loss rate and (b) error rate in communication testing.

USB and Ethernet. Meanwhile, the communication latency of USB is significantly lower than that of Ethernet and serial ports, demonstrating the fast response advantage of USB in communication.

The packet loss rate and error rate of communication testing are shown in Fig. 13. Fig. 13(a) showed the packet loss rate test results, with Ethernet packet loss rate below 0.3%, serial port packet loss rate below 0.05%, and USB packet loss rate

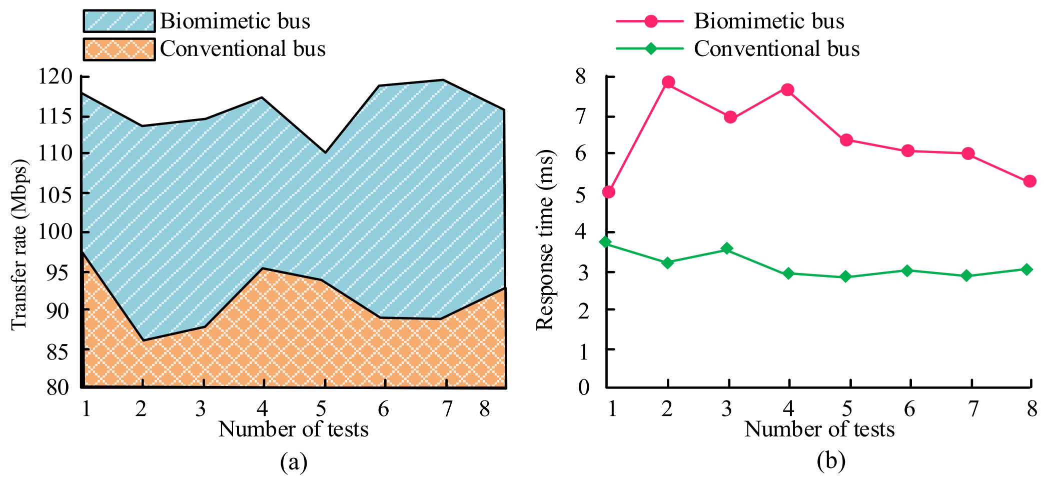

Fig. 14. Biomimetic bus test results. (a) Transfer rate. (b) Response time.

below 0.01%. Fig. 13(b) shows the bit error rate test results. The Ethernet bit error rate was below 1×10^{-12}, the serial port bit error rate was below 1×10^{-5}, and the USB bit error rate was below 1×10^{-10}. The results show that the communication packet loss rate and error rate of Ethernet, serial port, and USB are all low, indicating that their communication stability is high and can meet the communication quality requirements of the bird repellent system.

The results of the biomimetic bus test are shown in Fig. 14. Fig. 14(a) shows the data transmission rate. The biomimetic bus data transmission rates were all higher than 110 Mb/s, with an average of 116.3 Mb/s. The data transmission rate of conventional buses was below 100 Mb/s, with an average of 91.2 Mb/s. Fig. 14(b) shows the response time. Under the biomimetic bus, the response time was below 4 ms, with an average of 3.2 ms. Under conventional bus, the response time was over 5 ms, with an average of 6.4 ms. The results indicate that the biomimetic bus outperforms conventional buses in terms of data transmission rate and response time, demonstrating higher performance.

Compared with the baseline task allocation strategy (first-come–first-served), the proposed dynamic task allocation strategy showed significant advantages in terms of task completion time and resource utilization. The comparison results are shown in Table I. In the same bird drive task scenario, the average task completion time of dynamic task allocation strategy is shortened by 12.2 ms, and the device resource utilization is increased by 18.4%. This shows that the dynamic task allocation strategy can be flexibly adjusted according to the urgency of the task and the resource status of the equipment, so as to complete the bird drive task more effectively.

D. On-Site Testing Results and Discussion

In order to verify the bird repellent effect of the system in actual environments, ensure its stable and efficient functioning, protect the target area from bird invasion, and collect data to provide a basis for subsequent optimization, on-site testing is conducted in typical areas where birds frequently appear. Behavioral data, such as the number and duration of bird stays in different areas, are continuously observed and recorded to evaluate the performance of the bird repellent system. The on-site testing diagram of the bird repellent system is shown in Fig. 15.

TABLE I
RESULTS COMPARED TO BASELINE TASK ALLOCATION STRATEGIES

Strategies	Average task completion time (ms)	Equipment resource utilization (%)
Dynamic task assignment	3.5	85.1
First-come-first-service	15.7	66.7

Fig. 15. On-site testing diagram of the bird repellent system.

The on-site test results of the bird repellent system are shown in Fig. 16. Fig. 16(a) shows the results of bird repellent efficiency testing. After 48 h of on-site testing, the bird repellent efficiency reached 94.6%, effectively reducing the interference of birds in the airport area. Fig. 16(b) shows the stability test results of the system. Within 48 h of continuous operation, only two faults occurred, and the recovery time was

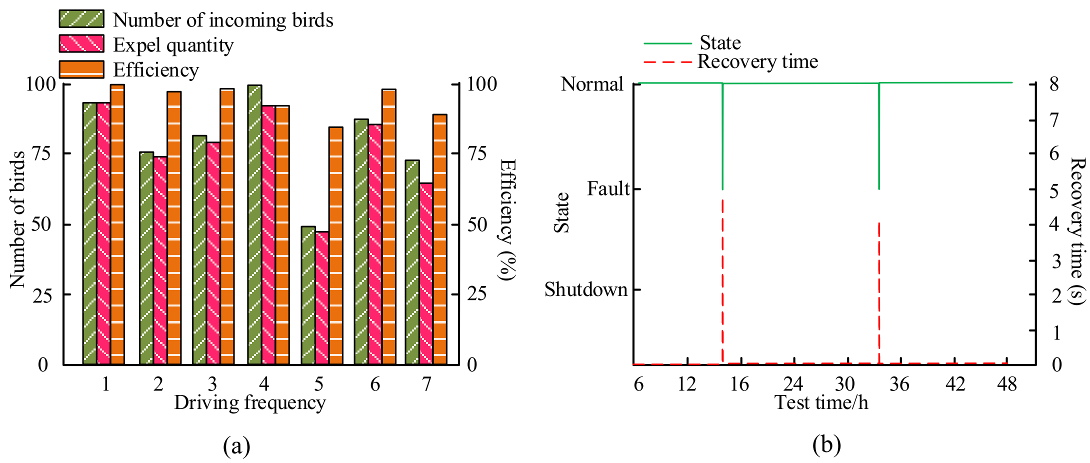

Fig. 16. On-site testing results of the bird repellent system. (a) Bird repelling efficiency. (b) System stability.

TABLE II
RESULTS OF COMPARISON OF BIRD REPELLENT SYSTEMS

Bird repellent system	Bird classification recognition accuracy (%)	Rate of error in assessment of bird hazard (%)
Research proposed	95.2	7.6
Ref [43]	76	41
Ref [44]	92	-
Ref [45]	92.3	-

TABLE III
PRACTICAL TEST RESULTS IN AGRICULTURAL FIELDS AND INDUSTRIAL SITES

Experimental scene	**Bird rejection rate (%)**	**Bird detection accuracy (%)**	**Response time (ms)**
Agricultural fields	92.8	94.3	165
Industrial sites	93.5	95.2	158

less than 5 s. At the same time, the bird repellent system has strong adaptability to the environment.

Under different weather and lighting conditions, the system performance fluctuates slightly and can work normally.

In order to verify the advantages of the bird repellent system proposed in the study, the experiment compares the existing bird repellent system based on traditional methods [43], [44], [45], and the results are shown in Table II. Compared with the existing system, the proposed work improves the accuracy of bird classification recognition by 3.2% and reduces the error rate of flock risk assessment by 33.4%. This shows that the fuzzy matching algorithm has significant advantages in processing complex bird information.

In order to verify the effectiveness of the system, practical tests were carried out in the agricultural field and in industrial sites, and the results are shown in Table III. In the field of agriculture, the rejection rate of birds reached 92.8%, the bird detection accuracy rate was 94.3%, and the system response time was 165 ms. In industrial sites, the system rejection rate of birds was 93.5%, the bird detection accuracy was 95.2%, and the system response time was 158 ms. These quantitative results show that the proposed system can effectively achieve bird repellent function in different environments.

Compared with existing bird repellent systems, the system proposed in this study has significant advantages in key performance indicators, such as adaptability, efficiency, and scalability. In terms of adaptability, the system can cope with complex bird behavior and changing scenarios in different environments. In terms of efficiency, the system significantly improves the completion efficiency of the bird drive task through dynamic task allocation strategy and optimized communication protocol. In terms of scalability, the modular design makes it easy to add new functional modules or upgrade existing functions. For example, in the field of agriculture, the system can automatically adjust the bird drive strategy according to seasonal changes and differences in bird species, effectively improving the efficiency of bird drive.

V. CONCLUSION

In response to the software design issues of the bird repellent system, the study introduced modular and pattern-based methods, proposing a system framework, functional modules, dynamic task allocation strategy, system software operation process, and pattern-based biomimetic bus design. The entire research was completed by combining experimental environment construction and module testing. The experimental results showed that the accuracy of the perception module reached 93.5%, and the response time tended to be 215 ms.

When the data volume was 1000, the processing time of the information processing module was about 2567 ms. After the image acquisition quality was stable, the signal-to-noise ratio during the daytime and nighttime was about 43.6 and 40.5 dB, respectively. Within 500 m^2, the success rate of the driving module was about 97.6% during the daytime and about 94.2% at nighttime, but the success rate decreased when the driving range exceeded 950 m^2. The error rate of human–machine interaction module status detection was less than 2.6%, and the remote-control response time tended to be 157 ms.

In communication testing, Ethernet, serial port, and USB communication had high stability, and the biomimetic bus was superior to conventional buses in terms of data transmission speed and response time. In on-site testing, the bird repellent efficiency reached 94.6%, with only two malfunctions within 48 h and a recovery time of less than 5 s. This bird repellent system can effectively reduce the interference of birds in airport areas and has strong stability.

The bird repellent system proposed in this study not only has significant application value in the field of bird repellent but also can play an important role in the field of wildlife monitoring and pest control. In wildlife monitoring, the system can provide data support for wildlife protection and research through real-time monitoring and analysis of animal behavior. In the field of pest control, the system can accurately identify and locate pests, and achieve pest control through corresponding drive or kill measures to improve agricultural production efficiency.

The modular and pattern-based design method adopted in this study is not only applicable to the bird repellent system but also can be widely applied to other IoT applications. Modular design enables the system to flexibly combine and expand function modules according to different application scenarios, improving the maintainability and expansibility of the system. The pattern-based approach can effectively deal with the complex and changeable data information in IoT application, and improve the intelligent level of the system. For example, in the smart home system, new devices and functional modules can be easily added through modular design, and the pattern-based approach can realize the intelligent analysis of home environmental data and the automatic control of equipment.

In the implementation of the system, there are potential challenges, such as hardware limitations, environmental factors, and cost considerations. In response to hardware limitations, hardware configuration can be optimized, and high-performance sensors and processors can be used to ensure stable operation of the system. For environmental factors, the system can adopt waterproof, dustproof, and other protective measures, and adopt adaptive algorithms to cope with bird behavior changes under different environmental conditions. In terms of cost, through reasonable selection of hardware components and optimization of system architecture, the overall cost of the system is reduced, so that it has a high cost performance.

The limitation of the study is that it does not fully consider the impact of all complex environmental factors on the system. Future research needs to further optimize the performance of the system in extremely complex environments.

Acknowledgment

Stephen Eta Zhou would like to thank his past self when he began self-learning programming and technology from scratch, gradually accumulating knowledge and experience without the guidance of a direct mentor or assistant. He also want to thank the Internet, books, and open-source communities. They have not only allowed him to rapidly access cutting-edge information and technologies but have also provided invaluable insights and inspirations during his thinking and practice. Finally, he sincerely wish that his insights and achievements can assist those who, like him, hold dreams but lack guidance, providing them with support and strength for their growth, especially for those who feel lost along the way.

References

[1] M. Mafarja et al., "An efficient high-dimensional feature selection approach driven by enhanced multi-strategy grey wolf optimizer for biological data classification," *Neural Comput. Appl.*, vol. 35, no. 2, pp. 1749–1775, 2023, doi: 10.1007/s00521-022-07836-8.

[2] A. Bajpai, T. P. G. Kumar, G. Sreenivasan, and S. K. Srivastav, "System design, automatic data collection framework and embedded software development of Internet of Things (IoT) for air pollution monitoring of Nagpur metropolis," *J. Indian Soc. Remote Sens.*, vol. 52, no. 10, pp. 2347–2359, 2024, doi: 10.1007/s12524-024-01943-w.

[3] A. M. Stuart et al., "Alternative domestic rodent pest management approaches to address the hazardous use of metal phosphides in low-and middle-income countries," *J. Pest Sci.*, vol. 98, pp. 89–111, Jan. 2025.

[4] A. F. El-Sayed, *Bird Strike in Aviation: Statistics, Analysis and Management*. Hoboken, NJ, USA: Wiley, 2019.

[5] S. M. Govorushko, "Mammals and birds as agricultural pests: A global situation," *Agric. Biol.*, no. 6, pp. 15–25, 2014.

[6] J. Astill, R. A. Dara, E. D. G. Fraser, B. Roberts, and S. Sharif, "Smart poultry management: Smart sensors, big data, and the Internet of Things," *Comput. Electron. Agric.*, vol. 170, Mar. 2020, Art. no. 105291.

[7] K. U. Sarker, R. Hasan, A. B. Deraman, and S. Mahmmod, "A distributed software project management framework," *J. Adv. Inf. Technol.*, vol. 14, no. 4, pp. 685–693, 2023, doi: 10.12720/jait.14.4.685-693.

[8] A. R. Tanner et al., "Epicosm—A framework for linking online social media in epidemiological cohorts," *Int. J. Epidemiol.*, vol. 52, no. 3, pp. 952–957, Jun. 2023, doi: 10.1093/ije/dyad020.

[9] T. Couppey, L. Regnacq, R. Giraud, O. Romain, Y. Bornat, and F. Kolbl, "NRV: An open framework for in silico evaluation of peripheral nerve electrical stimulation strategies," *PLoS Comput. Biol.*, vol. 20, no. 7, Jul. 2024, Art. no. e1011826, doi: 10.1371/journal.pcbi.1011826.

[10] Y. Cho et al., "A study on bird deterrent system to improve the performance of repelling harmful birds," *J. Korean Soc. Manuf. Process Eng.*, vol. 19, no. 8, pp. 15–21, 2020.

[11] X. Li and J. Yu, "Joint attention mechanism for the design of anti-bird collision accident detection system," *Electron. Res. Archive*, vol. 30, no. 12, pp. 4401–4415, 2022.

[12] E. Aman and H.-C. Wang, "A deep learning-based embedded system for pest bird sound detection and proximity estimation," *Eur. J. Eng. Technol. Res.*, vol. 9, no. 1, pp. 53–59, 2024.

[13] Y.-C. Chen, J.-F. Chu, K.-W. Hsieh, T.-H. Lin, P.-Z. Chang, and Y.-C. Tsai, "Automatic wild bird repellent system that is based on deep-learning-based wild bird detection and integrated with a laser rotation mechanism," *Sci. Rep.*, vol. 14, no. 1, 2024, Art. no. 15924.

[14] T. Faschang and G. Macher, "An open software-based framework for automotive cybersecurity testing," in *Proc. 30th Eur. Syst., Softw. Services Process Improv.*, 2023, pp. 316–328, doi: 10.1007/978-3-031-42307-9_22.

[15] R. Alonso, R. E. Haber, F. Castaño, and D. R. Recupero, "Interoperable software platforms for introducing artificial intelligence components in manufacturing: A meta-framework for security and privacy," *Heliyon*, vol. 10, no. 4, Feb. 2024, Art. no. e26446, doi: 10.1016/j.heliyon.2024.e26446.

[16] M. Gemeno, "Developing conceptual framework of software defect prediction in software testing: The case of ethiopian software industries," *Int. J. Technol. Syst.*, vol. 8, no. 1, pp. 1–13, 2023, doi: 10.47604/ijts.1834.

[17] S. Sha, C. Li, X. Wang, Z. Wang, and Y. Luo, "Hardware-software collaborative tiered-memory management framework for virtualization," *ACM Trans. Comput. Syst.*, vol. 42, nos. 1–2, pp. 1–32, 2024, doi: 10.1145/3639564.

[18] F. Di Marcantonio, M. Incudini, D. Tezza, and M. Grossi, "Quantum Advantage Seeker with Kernels (QuASK): A software framework to speed up the research in quantum machine learning," *Quantum Mach. Intell.*, vol. 5, no. 1, p. 20, 2023, doi: 10.1007/s42484-023-00107-2.

[19] D. Ajiga, P. A. Okeleke, S. O. Folorunsho, and C. Ezeigweneme, "Methodologies for developing scalable software frameworks that support growing business needs," *Int. J. Manag. Entrep. Res.*, vol. 6, no. 8, pp. 2661–2683, 2024.

[20] G. Bianchini and P. Sánchez-Baracaldo, "TreeViewer: Flexible, modular software to visualise and manipulate phylogenetic trees," *Ecol. Evol.*, vol. 14, no. 2, Feb. 2024, Art. no. e10873, doi: 10.1002/ece3.10873.

[21] M. Picone, M. Mamei, and F. Zambonelli, "A flexible and modular architecture for edge digital twin: implementation and evaluation," *ACM Trans. Internet Things*, vol. 4, no. 1, pp. 1–32, 2023, doi: 10.1145/3573206.

[22] K. Farias and L. Lazzari, "Event-driven architecture and REST architectural style: An exploratory study on modularity," *J. Appl. Res. Technol.*, vol. 21, no. 3, pp. 338–351, 2023, doi: 10.22201/icat.24486736e.2023.21.3.1764.

[23] R. R. Althar, D. Samanta, S. Purushotham, S. S. Sengar, and C. Hewage, "Design and development of artificial intelligence knowledge processing system for optimizing security of software system," *SN Comput. Sci.*, vol. 4, no. 4, p. 331, 2023, doi: 10.1007/s42979-023-01785-2.

[24] B. Arasteh, "Clustered design-model generation from a program source code using chaos-based metaheuristic algorithms," *Neural Comput. Appl.*, vol. 35, no. 4, pp. 3283–3305, 2023, doi: 10.1007/s00521-022-07781-6.

[25] X. Zhaoguo, Z. Zhenhua, and W. Yi, "Performance assessment and optimization of bird prevention devices for transmission lines based on Internet of Things technology," *Appl. Math. Nonlin. Sci.*, vol. 9, no. 1, pp. 1–16, 2024, doi: 10.2478/amns-2024-3408.

[26] S. Phetyawa, C. Kamyod, T. Yooyatiwong, and C. G. Kim, "Application and challenges of an IoT bird repeller system as a result of bird behavior," in *Proc. 25th Int. Symp. Wireless Pers. Multimedia Commun. (WPMC)*, 2022, pp. 323–327.

[27] D. Adami, M. O. Ojo, and S. Giordano, "Design, development and evaluation of an intelligent animal repelling system for crop protection based on embedded edge-AI," *IEEE Access*, vol. 9, pp. 132125–132139, 2021.

[28] I. S. Abeywardena, "OXREF: Open XR for education framework," *Int. Rev. Res. Open Distrib. Learn.*, vol. 24, no. 3, pp. 185–206, 2023, doi: 10.19173/irrodl.v24i3.7109.

[29] B. Tang, V. K. Shah, V. Marojevic, and J. H. Reed, "AI testing framework for next-G O-RAN networks: Requirements, design, and research opportunities," *IEEE Wireless Commun.*, vol. 30, no. 1, pp. 70–77, Feb. 2023, doi: 10.1109/mwc.001.2200213.

[30] M. G. Drouven, A. J. Caldéron, M. A. Zamarripa, and K. Beattie, "PARETO: An open-source produced water optimization framework," *Optim. Eng.*, vol. 24, no. 3, pp. 2229–2249, 2022, doi: 10.1007/s11081-022-09773-w.

[31] A. S. Allam, H. Bassioni, M. Ayoub, and W. Kamel, "A framework for obtaining a BIM-compatible design solution based on quantitative decisions on building performance," *Int. J. Constr. Manag.*, vol. 24, no. 11, pp. 1176–1190, 2022, doi: 10.1080/15623599.2022.2118103.

[32] M. Huss, D. R. Herber, and J. M. Borky, "Comparing measured agile software development metrics using an agile model-based software engineering approach versus scrum only," *Software*, vol. 2, no. 3, pp. 310–331, 2023, doi: 10.3390/software2030015.

[33] R. Di Felice et al., "A perspective on sustainable computational chemistry software development and integration," *J. Chem. Theory Comput.*, vol. 19, no. 20, pp. 7056–7076, Oct. 2023, doi: 10.1021/acs.jctc.3c00419.

[34] Y. Zhang et al., "Characterizing and detecting WebAssembly runtime bugs," *ACM Trans. Softw. Eng. Methodol.*, vol. 33, no. 2, pp. 1–29, 2023, doi: 10.1145/3624743.

[35] S. Pargaonkar, "A comprehensive research analysis of software development life cycle (SDLC) agile & waterfall model advantages, disadvantages, and application suitability in software quality engineering," *Int. J. Sci. Res. Publ.*, vol. 13, no. 8, pp. 120–124, 2023, doi: 10.29322/ijsrp.13.08.2023.p14015.

[36] M. M. Morovati, A. Nikanjam, F. Tambon, F. Khomh, and Z. M. Jiang, "Bug characterization in machine learning-based systems," *Empir. Softw. Eng.*, vol. 29, no. 1, p. 14, 2024, doi: 10.1007/s10664-023-10400-0.

[37] M. Bearman, J. H. Nieminen, and R. Ajjawi, "Designing assessment in a digital world: An organising framework," *Assess. Eval. Higher Educ.*, vol. 48, no. 3, pp. 291–304, 2022, doi: 10.1080/02602938.2022.2069674.

[38] H. O. Demirel, M. H. Goldstein, X. Li, and Z. Sha, "Human-centered generative design framework: An early design framework to support concept creation and evaluation," *Int. J. Human–Comput. Interact.*, vol. 40, no. 4, pp. 933–944, 2023, doi: 10.1080/10447318.2023.2171489.

[39] E. E. K. Senoo, E. Akansah, I. Mendonça, and M. Aritsugi, "Monitoring and control framework for IoT, implemented for smart agriculture," *Sensors*, vol. 23, no. 5, p. 2714, Mar. 2023, doi: 10.3390/s23052714.

[40] P. Szmeja et al., "ASSIST-IoT: A modular implementation of a reference architecture for the next generation Internet of Things," *Electronics*, vol. 12, no. 4, p. 854, 2023, doi: 10.3390/electronics12040854.

[41] N. D. Nguyen et al., "Towards designing a generic and comprehensive deep reinforcement learning framework," *Appl. Intell.*, vol. 53, no. 3, pp. 2967–2988, 2022, doi: 10.1007/s10489-022-03550-z.

[42] C. Shimizu, K. Hammar, and P. Hitzler, "Modular ontology modeling," *Semant. Web*, vol. 14, no. 3, pp. 459–489, 2023, doi: 10.3233/sw-222886.

[43] M. Arowolo, A. Adekunle, and J. Ade-Omowaye, "A real time image processing bird repellent system using raspberry Pi," *FUOYE J. Eng. Technol.*, vol. 5, pp. 1076–1082, Sep. 2020, doi: 10.46792/fuoyejet.v5i2.496.

[44] Q. Chen et al., "An experimental study of acoustic bird repellents for reducing bird encroachment in pear orchards," *Front. Plant Sci.*, vol. 15, Sep. 2024, Art. no. 1365275.

[45] A. Mirugwe, J. Nyirenda, and E. Dufourq, "Automating bird detection based on webcam captured images using deep learning," in *Proc. 43rd Conf. South African Inst. Comput. Sci. Inf. Technol.*, 2022, pp. 62–76.

Stephen Eta Zhou received systematic training through open-education programs, including the MIT OpenCourseWare series, and has accumulated over ten years of hands-on experience in Linux kernel development and open-source community projects. Although not conferred with a formal university degree in computer science, he has completed numerous massive open online courses (MOOCs) and has published technical articles in peer-reviewed venues. He is an independent programmer and researcher from the City of Shanghai, China. He is a Software Department Manager with ShangHaiDieCheng-BM Software Unit, Shanghai. His current research interests include operating system kernel design, advanced system architecture, multicamera synchronization, and software framework design.

Neelam Mughees received the Ph.D. degree in electrical engineering from COMSATS University Islamabad, Islamabad, Pakistan, in 2024.

She is currently a Lecturer with National Textile University, Faisalabad, Pakistan. Her research interests include applications of machine learning, reinforcement learning, and deep learning models.